Outset · AI in Healthcare · Day 3 · Group 1

Picking good CRISPR guides before the lab

Given the 20 letters of a guide RNA, can a model tell -- before any lab work -- whether CRISPR-Cas9 will actually cut?

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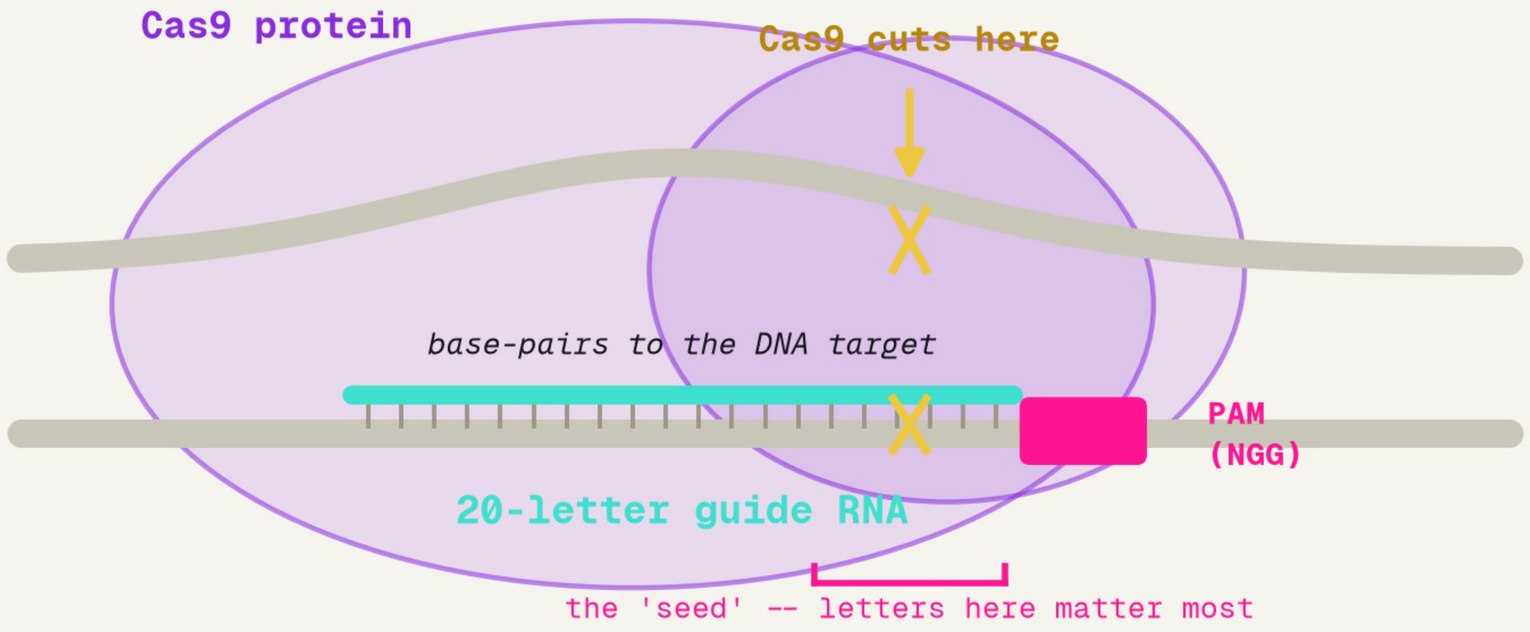
Overview

BACKGROUND

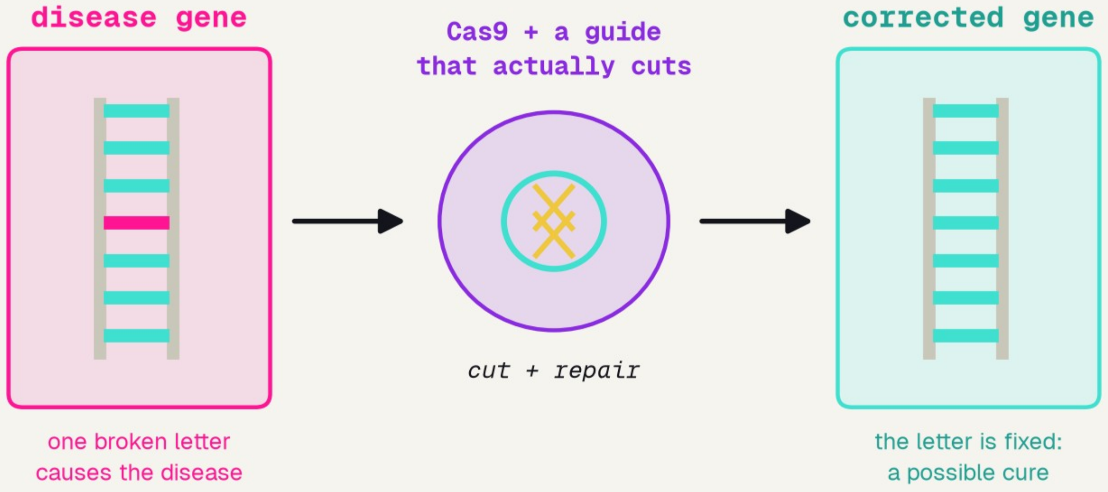
Molecular scissors, aimed by 20 letters



adapted from Hsu, Lander & Zhang 2014

A 20-letter guide RNA steers Cas9 to one spot in DNA; it cuts about three letters from the PAM -- and the letters in the seed, next to the PAM, matter most.

CRISPR as therapy: fixing a broken gene



the same tool already treats sickle-cell disease (approved 2023) -- but only if the guide cuts

concept: Hsu, Lander & Zhang 2014

THE CLINICAL SETTING

Who

patients with a genetic disease -- sickle-cell today, many more in trials

Where

research labs and gene-therapy clinics deciding what to build

The decision

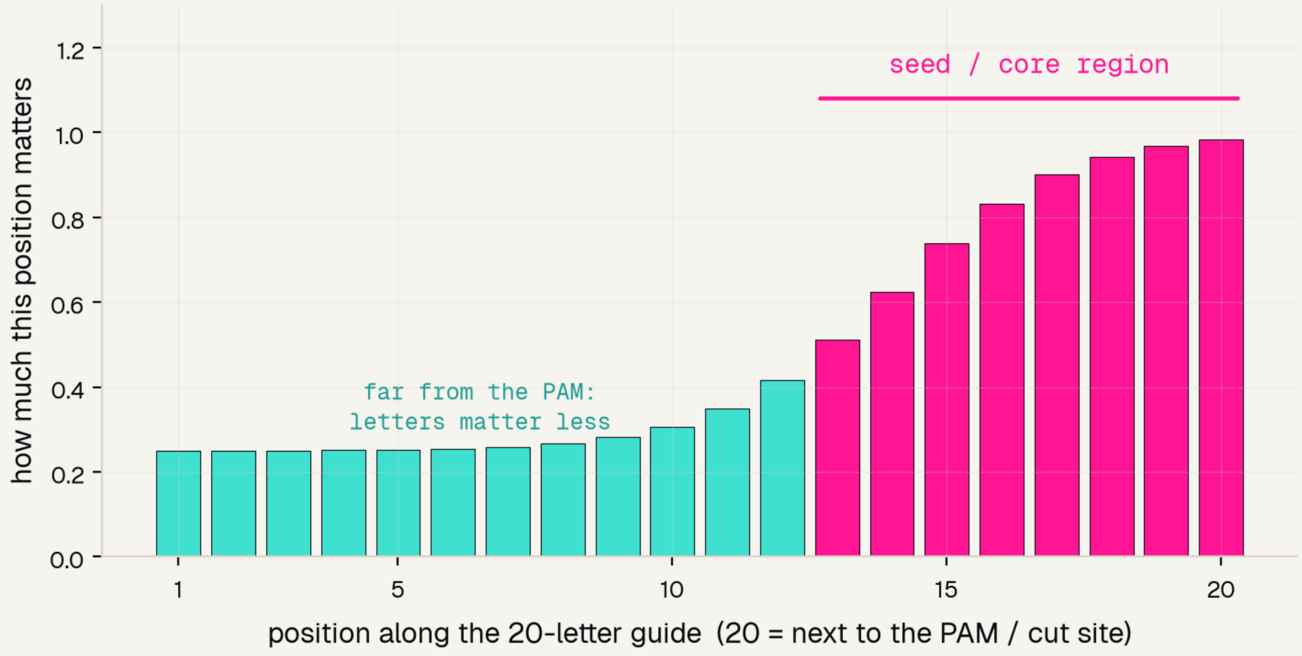
which 20-letter guide to make -- the one that will actually cut

Not all 20 letters count equally

THE ONE BIT OF BIOLOGY

Cas9 lands on the PAM, then reads the handful of letters right beside it -- the seed.

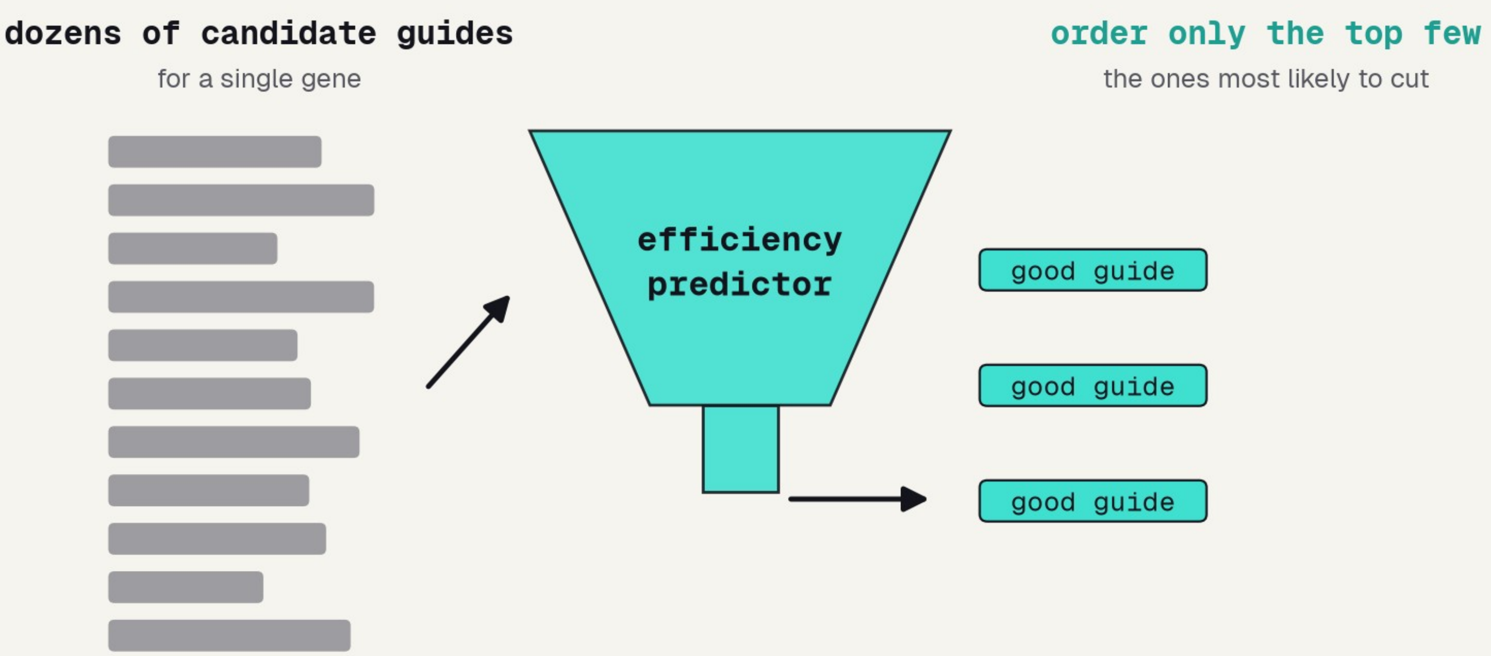
Mismatch a seed letter and the guide usually fails to cut. So the letters near the PAM should carry the most signal -- far more than the ones at the far end.



adapted from Zheng et al. 2017 / Doench et al. 2016

Remember this -- our model will rediscover it.

The catch: most candidate guides are duds



concept adapted from Doench et al. 2016

the old way

test dozens of guides in the lab, one by one -- weeks of work, and most turn out to be duds

the tool

score all of them from the letters in seconds, then order only the top few most likely to cut

Methodology

THE DATA

Why Doench 2016

5,310

guides, each ACTUALLY tested in the lab -- so every cut-score is a real measurement, not a simulation

17

genes targeted, a spread wide enough that the model learns sequence rules, not the quirks of one gene

the

benchmark: the dataset Doench built Rule Set 2 / Azimuth on -- what the whole field still compares against

The benchmark that set the standard

SET HONEST EXPECTATIONS

Doench 2016 built Rule Set 2 (Azimuth) on exactly these guides -- the state of the art.

Even so, its predicted scores only loosely track the measured ones (the cloud scatters around the line).

So our goal is to capture the real signal -- not to expect a perfect predictor.

adapted from Doench et al. 2016

THE MODEL

First, turn letters into numbers

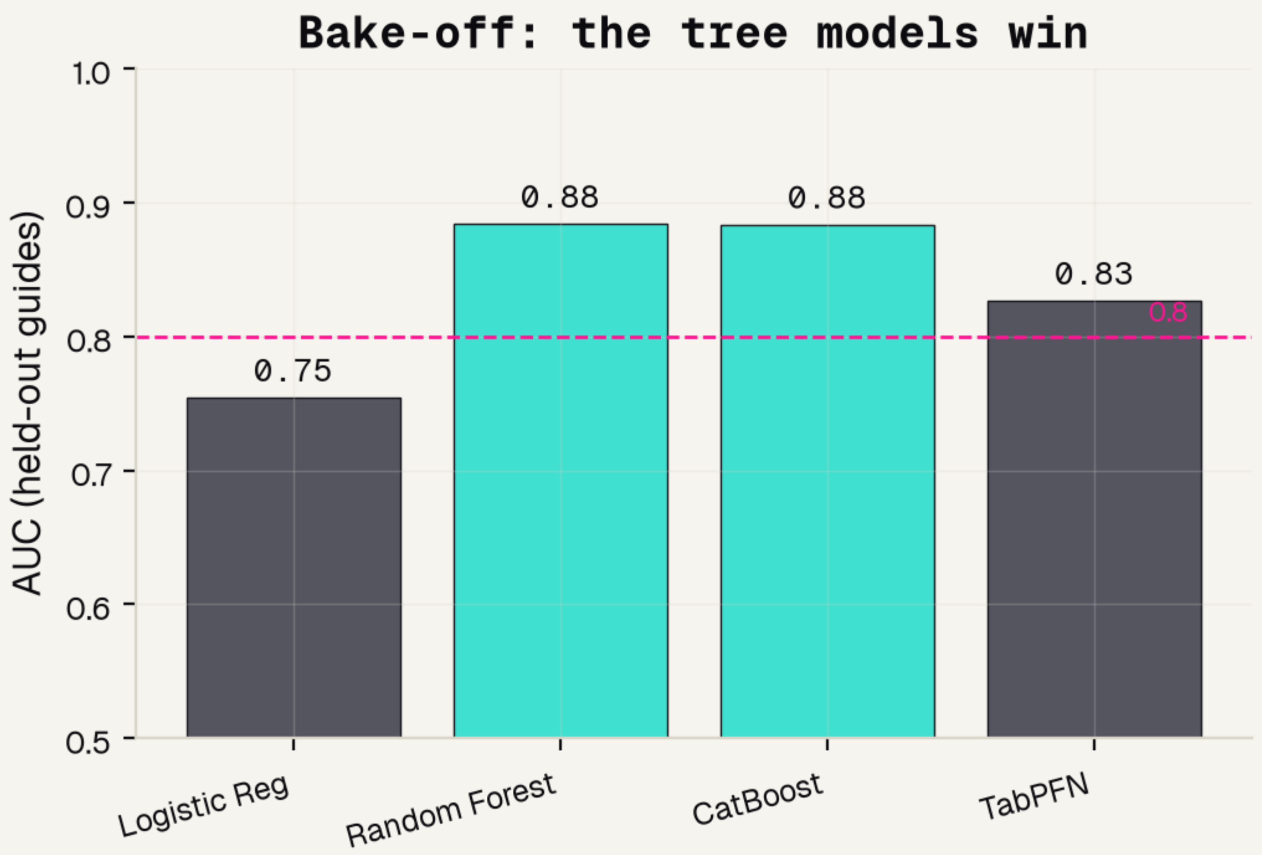
a computer cannot do math on the letter A -- so one-hot flips 4 switches per position:

	G	A	C	T	G	A	A	C	A	G
A		1				1	1		1	
C			1					1		
G	1				1					1
T				1						

One switch ON per column → 80 numbers that KEEP the order, so the seed near the PAM stays visible.

k-mer would just count letters and throw the order away -- so we use one-hot.

We tested four models - we did not guess



WE TESTED, WE DID NOT GUESS

0.88

the two TREE models (CatBoost and Random Forest) tie here -- both beat TabPFN (0.83) and crush logistic regression (0.75).

We pick CatBoost -- fast, strong, and reproducible.

Model and data processing

1	TASK	clearly-good (top-third efficiency) vs clearly-poor (bottom-third); the muddy middle is dropped
2	ENCODE	each 20-letter guide becomes an 80-number one-hot vector that keeps the position of every letter
3	SPLIT	a stratified 75% train / 25% test split -- every number we report is on guides the model never saw
4	MODEL	CatBoostClassifier, 600 trees, depth 6, learning rate 0.05, random_state 0 so it is reproducible
5	GRADE	AUC on the held-out test set -- not accuracy -- so a lazy one-class guess cannot game the score

Validation

RESULTS

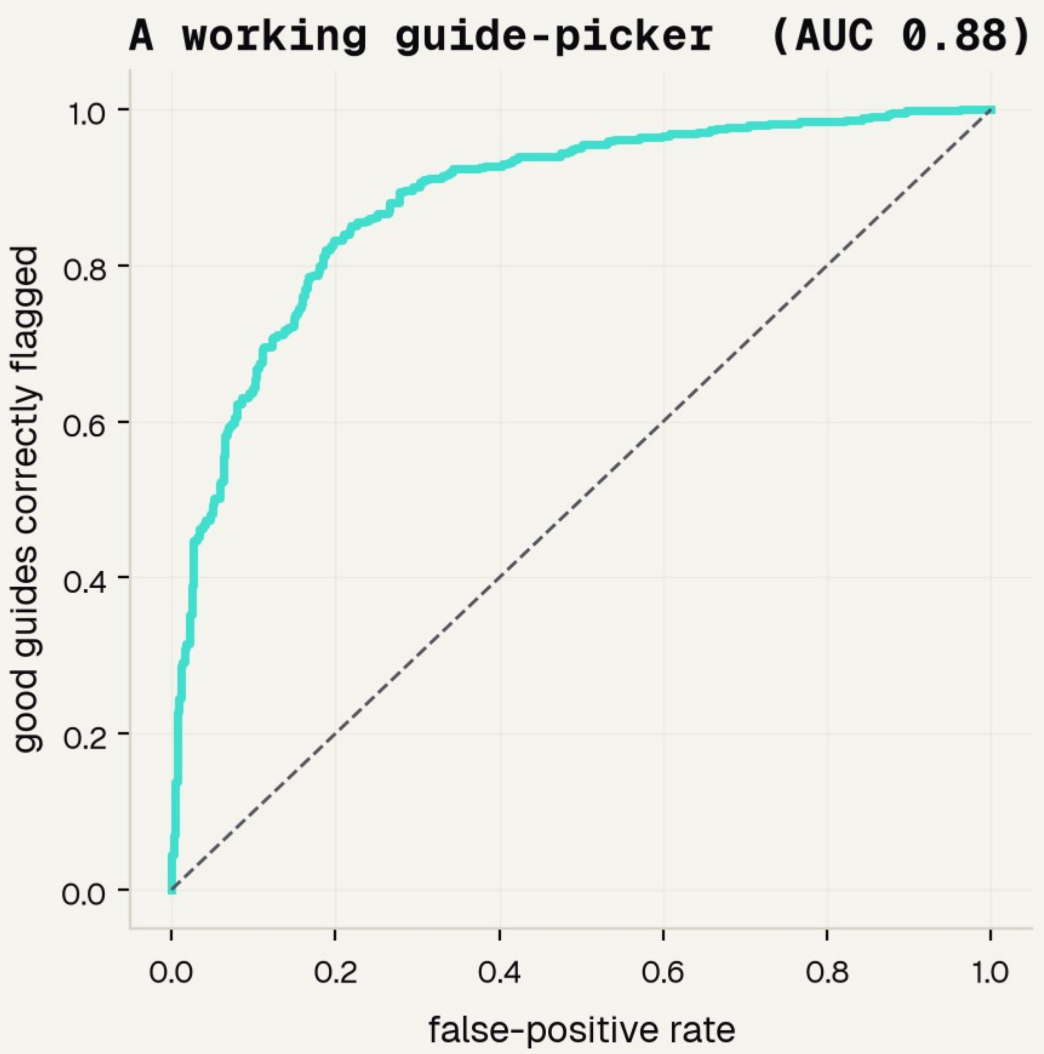
How we grade it: ROC and AUC

0.88

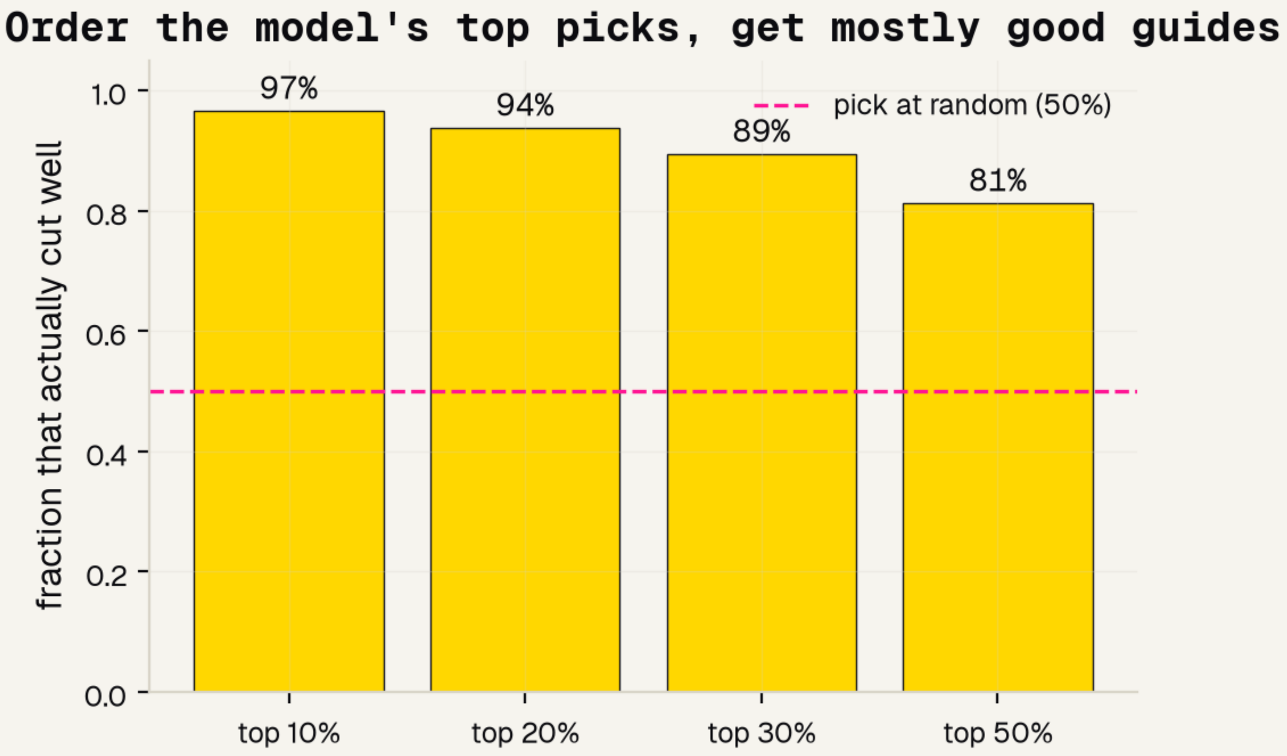
AUC -- the area under the ROC curve

The picker outputs a score, not a yes/no. The ROC curve sweeps every possible cut-off and plots good guides caught against false alarms; AUC is the area underneath.

0.5 = a coin flip   **1.0 = perfect**



The win: order its top picks

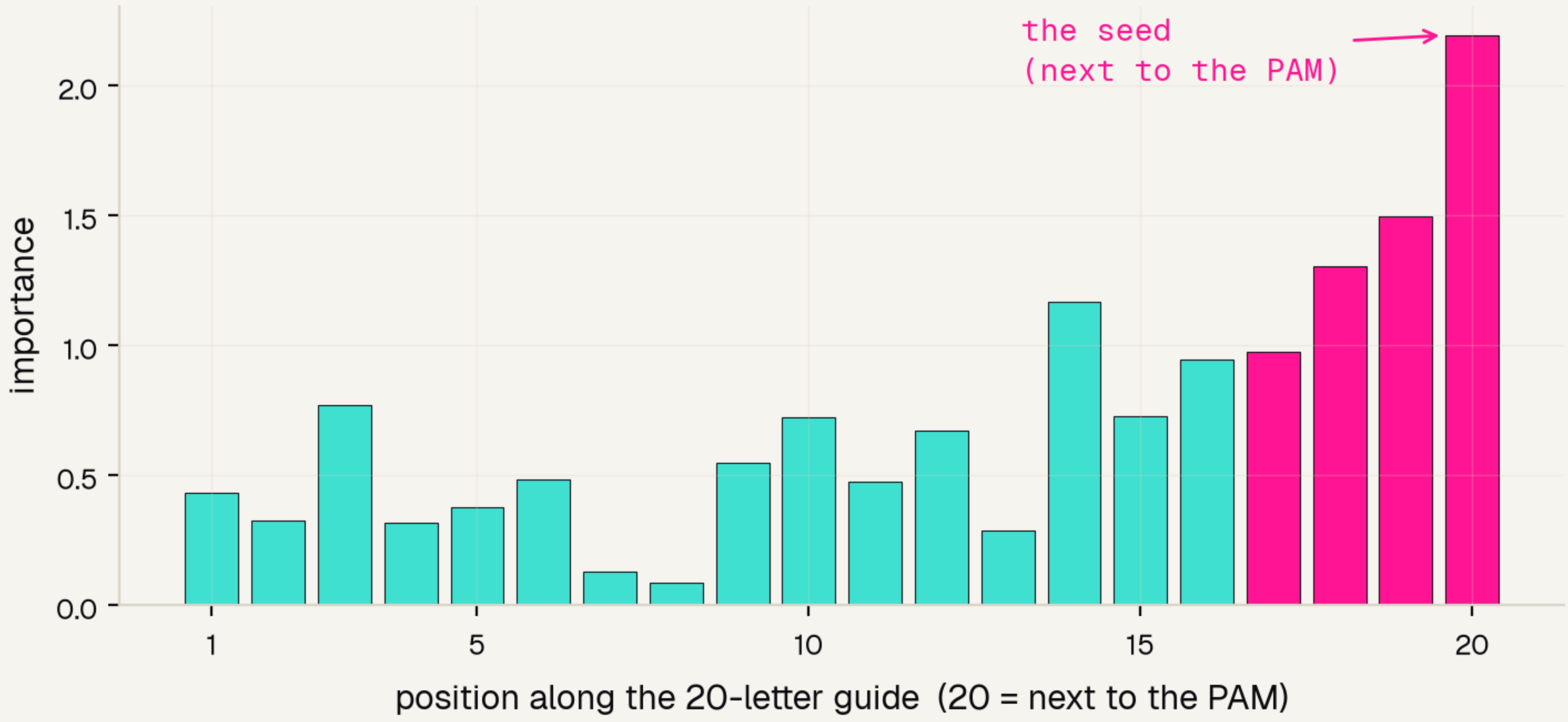


97% of the model's TOP 10% actually cut well

50% if you pick guides at random -- the baseline

weeks of lab time saved, because the shortlist is almost all winners

It rediscovered the biology



Feature importance = which positions the model leaned on. It peaks at position 20 -- right next to the PAM, the seed -- matching decades of CRISPR biology it was never told.

Fairness? There is nothing to audit - honestly

No patient. No demographics. Nothing to be unfair about.

Most medical-AI projects must check they work equally across sex, age, or skin tone. This one has no such groups -- the input is a 20-letter DNA sequence. Forcing a fake fairness analysis would be less honest than simply saying so.

The real equity question for CRISPR lives one step later -- in which patients a therapy is tested on and reaches -- outside this sequence model.

Overview

THE BOTTOM LINE

What it can and cannot do

IT WORKS

A real guide-picker on the field's benchmark: AUC 0.88, and its top-10% shortlist is 97% good guides -- real lab time saved.

AUC 0.88 · 97% precision

IT LEARNED REAL BIOLOGY

Importance peaks at the seed next to the PAM, matching decades of CRISPR science -- so we can trust WHY it works.

rediscovered the seed

BUT IT IS A DEMO

It cannot match a production tool, capture cell-type or chromatin effects, or say anything about off-target safety.

not a design tool -- yet

References

FOUNDATIONS & DATA

[1] Doench et al. 2016, Nat Biotechnol -- Rule Set 2 / Azimuth, and the dataset we use.

[2] Doench et al. 2014, Nat Biotechnol -- Rule Set 1: position + PAM predict cutting.

[3] Hsu, Lander & Zhang 2014, Cell -- how Cas9 is steered to a DNA site (the seed).

[4] Zheng et al. 2017, Sci Rep -- direct evidence the PAM-proximal seed matters most.

METHODS & REVIEWS

[5] Chuai et al. 2018, Genome Biol -- DeepCRISPR: deep learning for guide design.

[6] Kim et al. 2019, Sci Adv -- DeepSpCas9: more data + neural nets push accuracy further.

[7] Konstantakos et al. 2022, NAR -- a survey of guide-efficiency predictors.

[8] Abbaszadeh & Shahlai 2025, arXiv -- explainable AI + off-target safety.

The honest bottom line

A guide-picker earns its keep by shortlisting the winners -- and explaining itself.

Use it to build intuition and narrow the field -- then use a validated tool and the lab to actually choose and confirm a guide.

Thank You

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夢想